

Topic quiz: Projectile motion

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IBDP Physics SL · Grade 11 · A.1 Kinematics · 10 multiple-choice questions

Name: _____

Class: _____

Date: _____

How to use. Practise with the simulations and worksheets first. Then answer every question: circle one letter and write it in the grid below. Use $g = 9.81 \text{ m s}^{-2}$ and show any working next to the question.

Check yourself online: the same quiz with instant feedback and explanations —
<https://biba0112.github.io/physics-sims/quiz-projectile-motion/>



1	2	3	4	5	6	7	8	9	10

Question 1

A ball is dropped from a bench. At the same moment, an identical ball rolls horizontally off the same bench. Air resistance is negligible. Which ball reaches the floor first?

- ☐ (A) Both reach the floor at the same time
- ☐ (B) The dropped ball
- ☐ (C) The ball that rolls off
- ☐ (D) It depends on how fast the second ball rolls

Question 2

A stone is dropped from rest and falls freely for 2.00 s. How far does it fall?

- ☐ (A) 9.81 m
- ☐ (B) 19.6 m
- ☐ (C) 39.2 m
- ☐ (D) 4.91 m

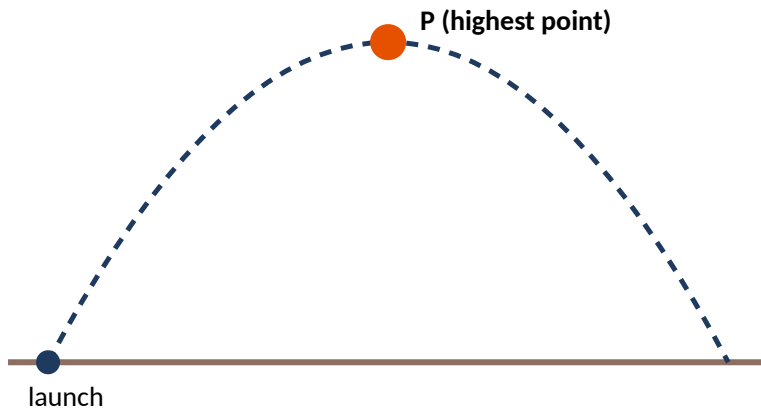
Question 3

A ball rolls off a horizontal bench 0.80 m high with a speed of 2.5 m s^{-1} . How far from the foot of the bench does it land?

- ☐ (A) 0.64 m
- ☐ (B) 0.71 m
- ☐ (C) 1.0 m
- ☐ (D) 2.0 m

Question 4

A ball is launched at an angle above level ground. Air resistance is negligible. Which statement about the ball at its highest point P is correct?



- (A) Its velocity is zero and its acceleration is zero
- (B) Its velocity is zero and its acceleration is 9.81 m s^{-2} downwards
- (C) Its velocity is horizontal and its acceleration is zero
- (D) Its velocity is horizontal and its acceleration is 9.81 m s^{-2} downwards

Question 5

A ball is kicked from level ground at 20.0 m s^{-1} at 30° above the horizontal. What is its maximum height?

- | | |
|------------|------------|
| (A) 5.10 m | (B) 10.2 m |
| (C) 15.3 m | (D) 20.4 m |

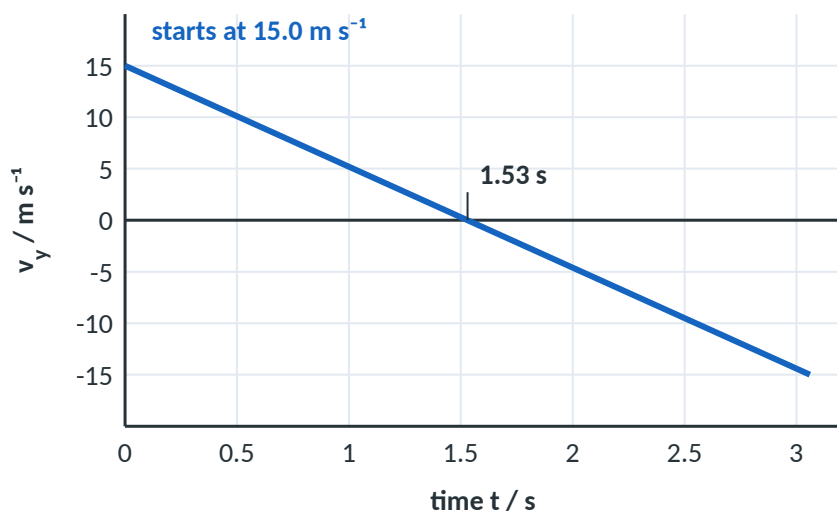
Question 6

At a certain speed, a ball launched from level ground at 30° lands 35 m away. At which other launch angle does the same speed give the same range?

- | | |
|----------------|----------------|
| (A) 15° | (B) 45° |
| (C) 60° | (D) 75° |

Question 7

The graph shows the vertical velocity v_y of a ball thrown straight upwards (up is positive). What is the maximum height the ball reaches above the launch point?



- (A) 0 m
- (B) 11.5 m
- (C) 15.0 m
- (D) 22.9 m

Question 8

A projectile moves without air resistance. What is the change in its velocity between $t = 0.5 \text{ s}$ and $t = 1.5 \text{ s}$?

- (A) Zero, because the horizontal velocity is constant
- (B) 9.81 m s^{-1} horizontally
- (C) It depends on the launch angle
- (D) 9.81 m s^{-1} vertically downwards

Question 9

A marble rolls horizontally off a table 1.25 m high. A cup on the floor is 1.50 m from the edge of the table. What launch speed lands the marble in the cup?

- (A) 1.20 m s^{-1}
- (B) 2.97 m s^{-1}
- (C) 4.20 m s^{-1}
- (D) 5.94 m s^{-1}

Question 10

Compared with its path in a vacuum, how does air resistance change the path of a thrown tennis ball?

- (A) Shorter range and lower maximum height; the ball comes down more steeply than it went up
- (B) Same range, but a lower maximum height
- (C) Shorter range and lower maximum height; the path is still symmetrical
- (D) Longer range, because the air supports the ball